31/3/26

Function transformer

This transformer is used to make the data into Normal distribution. Usually, the data's are not in normal distribution, in that case if we apply any ML algorithm, some of the ML algorithm output are not reliable (or) correct. Hence, we apply Function transformer to make the data Normally distributed to work on that

So, it is implemented by applying mathematical transforms to the dataset (comes under Feature engineering → Feature transformation → Mathematical transformation)

Types of Mathematical transformation :-

1) Log transform
2) Reciprocal transform
3) Power (sq / sqrt) transform
4) Box - cox transform
5) Yeo - Johnson transform.

Sklearn Library

- Function transformer
  - Log transform
  - Reciprocal transform
  - sq/sqrt transform
  - custom transform

  → Mostly work well with Linear / Logistic regression

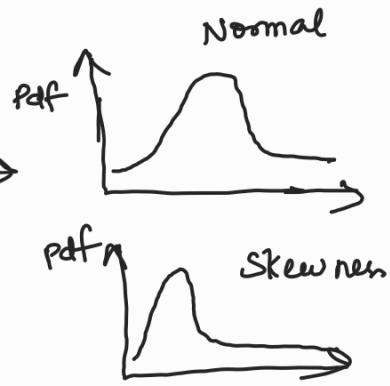
- Power transformer
  - Box-cox
  - Yeo-Johnson

- Quantile transformer

How to find the data is normally distributed (or) not ?

3 main visualization method

- Sns. plot (seaborn) → Gives the graph of Skewness/normal →



- pd.skew() (Pandas) = 0 (Normal)
  = −/+ values (skewed data)

- QQ plot → We have to note